# Case Study 1: Model Selection for Clustering

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# INTRODUCTION

Colorectal cancer(CRC) is one of the most diagnosed cancers that affects the colon or rectum part of the large intestine. Which is also known as bowel cancer because it affects the inner walls of the large intestine. The cancer occurs when cells in the rectum grow out of control and crowd out normal cells. The CRC tends to exhibit no symptoms until the later stages; early detection of the cancerous cells can help diagnose the disease at earlier stages. The methods used in the medical field to detect CRC are MRI, Colonoscopy, and CT scans of the Tumor region and the doctors analyze the images of this region carefully to analyze the cancerous tissues. Although there's been a lot of research into improving diagnostic and treatment methods. This needs expert analysis and resource consumption. Digital pathology has become a valuable tool in this field, there have been quite a number of improvements and research in machine learning algorithms can classify the tissue types. The classifying algorithms can analyse recorded images of the tissue patches and gain valuable insights into the disease's underlying structure.

## 1.1 Background

This study involves evaluating clustering performance of different unsupervised clustering techniques on the 5,000 non-overlapping image patches from hematoxylin & eosin (H&E) stained histological images of human colorectal cancer (CRC) and normal tissues in four different representations PathologyGAN, ResNet50, InceptionV3 and VGG16.

## 1.2 Clustering

Clustering is an unsupervised learning technique used to identify groups of similar entities within the unlabeled dataset and is widely used in biomedical applications where labeled datasets are often unavailable. It helps organize a large dataset into meaningful clusters enabling actionable insights for exploratory analysis. However, in this study, we worked with a labeled dataset of colorectal tissue patches classified into nine known histological classes. This allows us to validate the clustering performance quantitatively by comparing the predicted clusters against the ground truth classes. This study aims to analyze different clustering techniques to determine the most suitable algorithm for producing optimized group clusters for the given data set.

## 1.3 Dataset

The dataset consists of 5,000 colorectal tissue patches obtained from hematoxylin & eosin (H&E) stained histological images. The patches are represented as feature vectors which are extracted using four different convolutional neural network (CNN) models: PathologyGAN, VGG16, ResNet50, and InceptionV3. For this study, we focus on the feature vectors extracted from PathologyGAN and VGG16. These features have been reduced to 100 components using dimensionality reduction techniques, such as PCA and UMAP as it enhances the performance of clustering algorithms by retaining only the most relevant features of the data. Each data set is labeled with a tissue type from the available types Adipose (ADI), background (BACK), debris (DEB), lymphocytes (LYM), mucus (MUC), smooth muscle (MUS), normal colon mucosa (NORM), cancer-associated stroma (STR), colorectal adenocarcinoma epithelium (TUM)

## 1.4 Objective

This study evaluates the clustering performance of four major clustering algorithms - KMeans, Gaussian Mixture Models (GMM), DBSCAN and Louvain applied to a dataset of CRC tissue patches. The aim is to evaluate the clustering performance of each algorithm in grouping of similar tissue patches. The metrics Silhouette Scores and V-Measure are used to determine which clustering algorithm is best fit and organize the data into meaningful clusters by comparing against the given labels from the dataset.

# METHODOLOGY

## 2.1 Clustering Algorithms

This section explores four different unsupervised clustering techniques - K-Means, Gaussian Mixture Model (GMM), DBSCAN, and Louvain community detection to uncover patterns in the data. We discuss underlying algorithms and parameter selection for these algorithms to classify the data effectively.

### 2.1.1 K-Means

K-Means is a well-known partition clustering technique, where the data is partitioned into k unique clusters. It works by randomly selecting k initial centers, and assigning each data point to the nearest

center, and finally updating the centers to new values by taking the mean of all the cluster point dimension variables. The updated center values are used in the next iteration, until a convergence criteria is met.

**Parameter Selection:** The number of clusters, k, was determined using the elbow method, which helps identify the point where the within-cluster sum of squares (WCSS) begins to plateau, indicating the optimal number of clusters. To further refine the choice of k, a grid search was performed around the elbow point, testing values of k along with other relevant parameters to achieve well-separated clusters and optimize clustering performance.

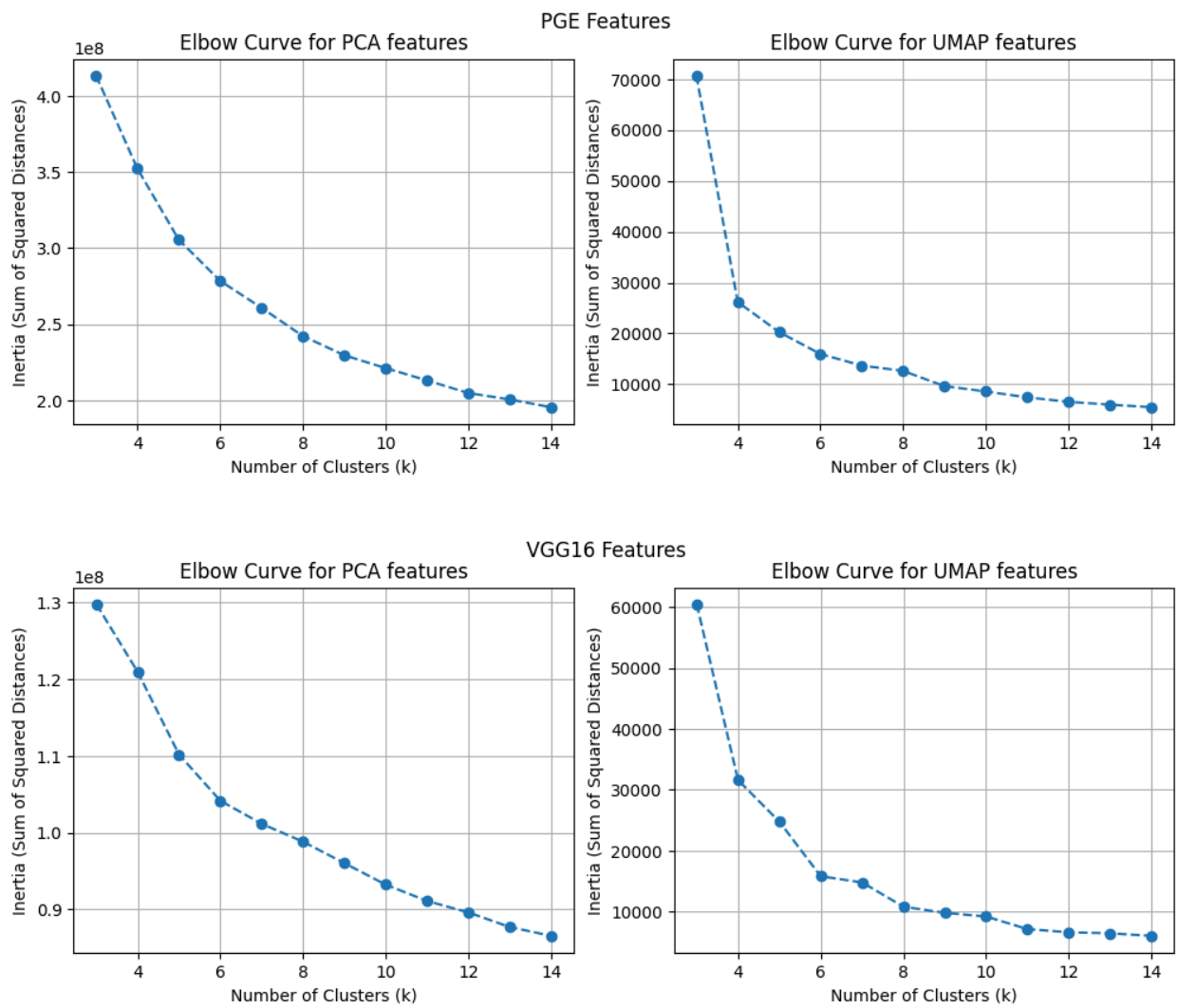


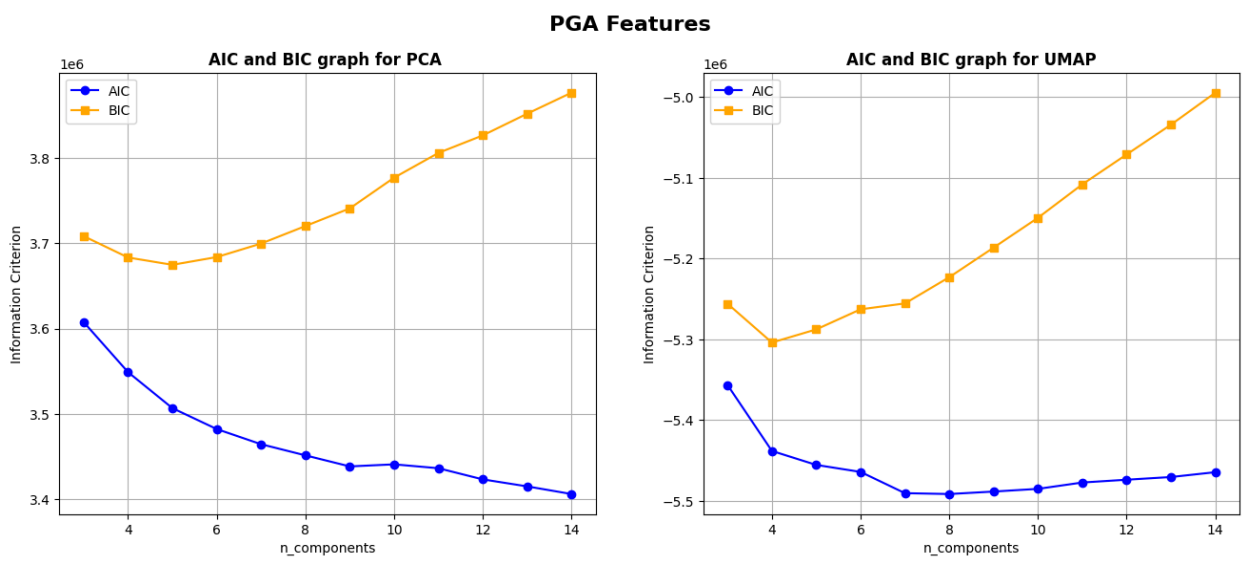

Figure 2.1.1.1: Elbow curve for PGE_PCA, PGE_UMAP, VGG16_PCA and VGG16_UMAP.

### 2.1.2 Gaussian Mixture Model(GMM)

The Gaussian Mixture Model(GMM) is a probabilistic clustering technique. The GMM uses Gaussian Probability distribution to find a mixture of multi-dimensional probability distribution models in the input dataset. Unlike Kmeans which is based on distance, the GMM model can be used to identify complex cluster shapes. So it is the best suit for any input dataset. Since GMM uses a probabilistic model it assumes each cluster follows a gaussian distribution defined by its mean and covariance matrix and provides probabilities of data points belonging to each cluster.

Gaussian initializes the parameters mean, covariance and weights randomly using a K-means or a random method. Calculates the probability of each point in the input data set to a Gaussian cluster using the parameters. Updates the parameter to maximize the likelihood of the observed data. Repeats the probability calculation and parameter updates until it converges or best likelihood of the observed data is achieved.

**Parameter Selection:** We used Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values to select the best likelihood that is achieved using the different number of components for a given feature set. We selected AIC values over BIC as it offers a better balance between model complexity and fit when dealing with high-dimensional dataset. To further refine the best component that matches the likelihood of our input data, we have performed a grid search on the range of best component values with different covariance types, iterations, and initial values to find the optimized clustering results. The grid search returns the best parameters and for the best n components.



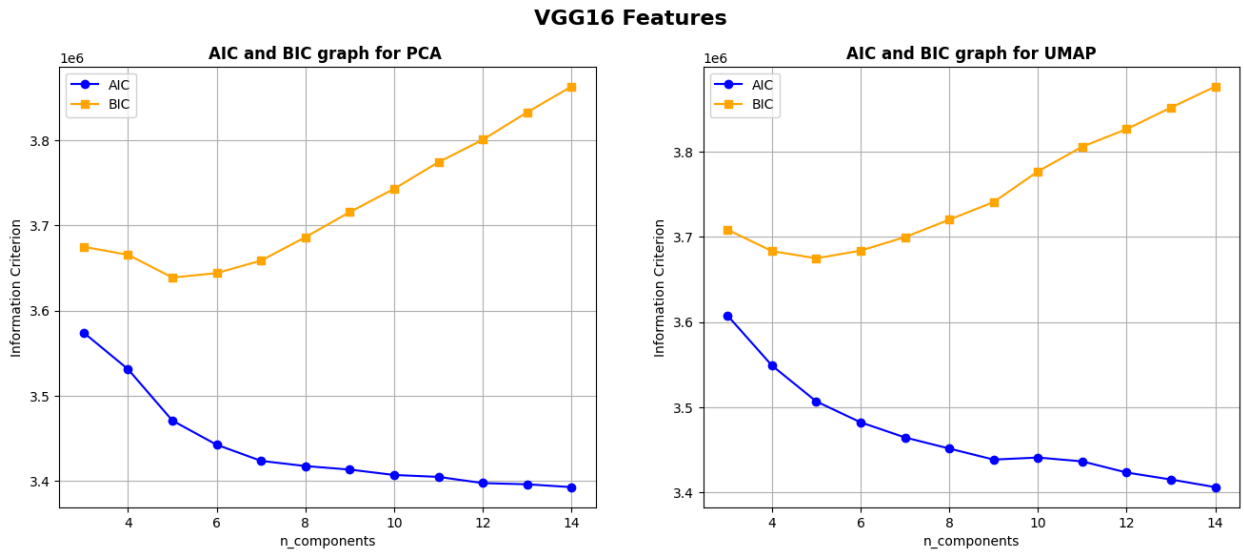



Figure 2.1.2.1: AIC and BIC plots for PGE_PCA, PGE_UMAP, VGG16_PCA and VGG16_UMAP

### 2.1.3 DBScan (Density-Based Spatial Clustering of Applications with Noise)

A density-based algorithm that identifies clusters as regions of high point density and labels sparse areas as noise. It arbitrarily selects a point 'p', retrieves all the points density reachable from it and continues the process until all the points have been processed. It is immune to the potential distortions in observations produced due to outliers, as it effectively prunes out the Outliers.

**Parameter Selection**: It is a common practice to set min_samples to at least twice the number of dimensions of the dataset, so we chose a range starting from 200, as our dataset has 100 dimensions. The optimal value of Epsilon (eps) is identified at the point of maximum curvature in the K-Distance Graph as shown below, which indicates the ideal eps for defining clusters. The range for eps was selected based on the elbow point observed in the k-distance graph. These parameters, along with other relevant settings, were used in a grid search to fine-tune DBScan.

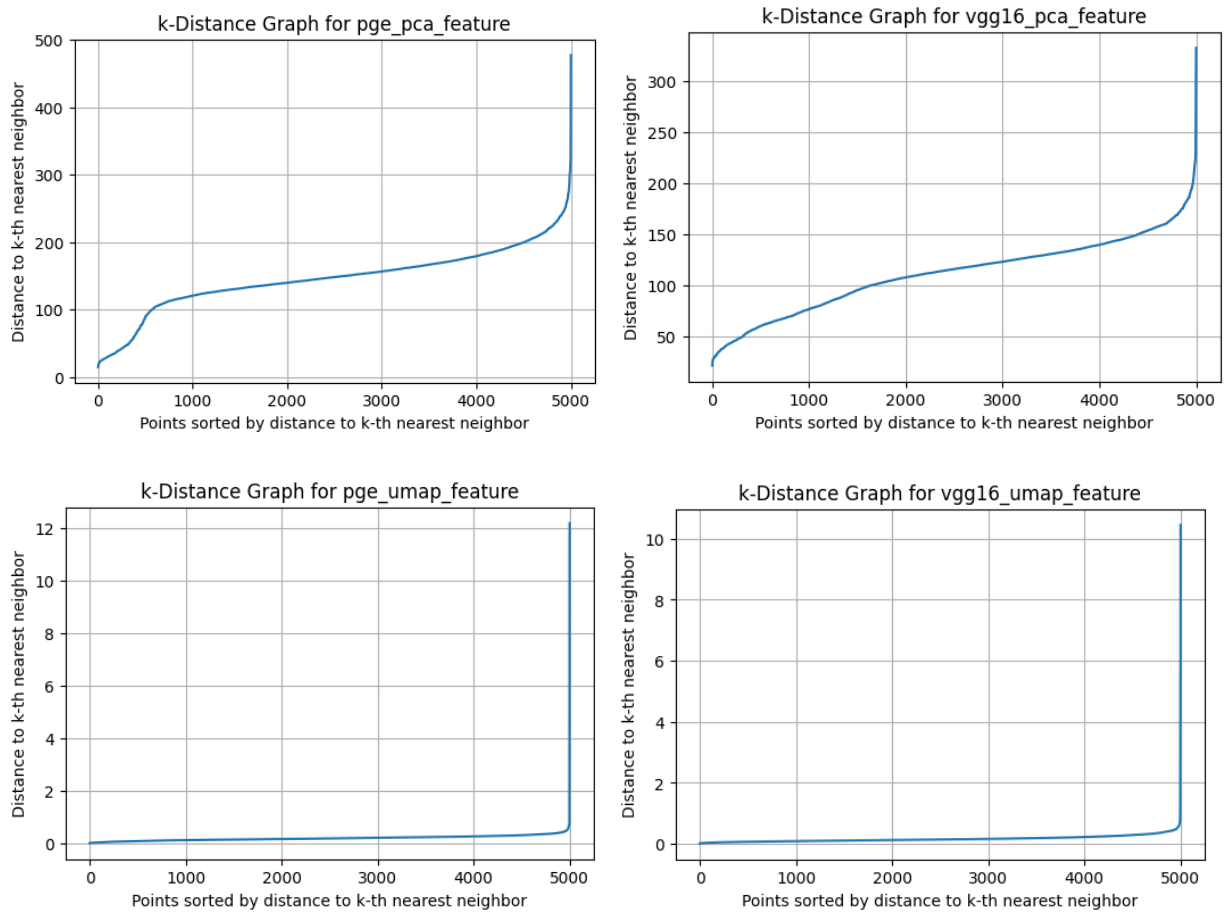



Figure 2.1.3.1: K-Distance Graph for PGE_PCA, PGE_UMAP, VGG16_PCA and VGG16_UMAP

### 2.1.4 Louvain community detection

The Louvain method for community detection is a greedy optimization method and it is non-deterministic by nature. It is based on the idea of optimizing Modularity. The formula for Modularity H is as shown in figure 2.1.4.1. Where, Kc is the sum of the degrees of the nodes, C is community and m is the total number of edges in the network. It works similar to hierarchical clustering, where each data point is considered a node and its own community in the initial run, singleton partition. As the algorithm iterates through each of the points and groups them into communities, such that the number of edges within communities is maximized while the edges between communities are minimized. In this report we are making use of the K-nearest neighbors (k-NN) algorithm to construct a K-NN graph. This graph is then used as an input to the modularity optimization. As the algorithm repeatedly merges communities the modularity score increases and the process stops when no further improvements in modularity can be

made. This is suited for large-scale networks and is used in various domains such as social networks, biology, and image segmentation.

$$\mathcal{H} = \frac{1}{2m} \sum_c (e_c - \gamma \frac{K_c^2}{2m}),$$

Figure 2.1.4.1: Formula of Modularity H

**Parameter Selection:** Resolution(γ) in Louvain clustering has a huge impact on the clustering result. When the resolution is less than 1, the algorithm will favor larger communities and when it's less than 1 it will favor smaller communities. This control on the parameters affects the granularity of the clustering. The figure 2.1.4.2 displays this effect on modularity and the number of clusters in PGE and VGG16 feature sets. Silhouette score and V-measure are a good tool to analyse the parameter selection process. They help in finding the best parameters to balance between obtaining meaningful clusters and avoiding overfitting or underfitting. Additionally, we can measure the entropy with the help of Cluster Size distribution. It measures the randomness. The figure 2.1.4.3 showcases the plot for cluster size for cluster labels. There is a sharp decline in the cluster size as the cluster label increases. But in the case of PGE_PCA and VGG16_PCA feature sets as there was a higher resolution this led to smaller clusters. And the opposite was true for UMAP feature sets.

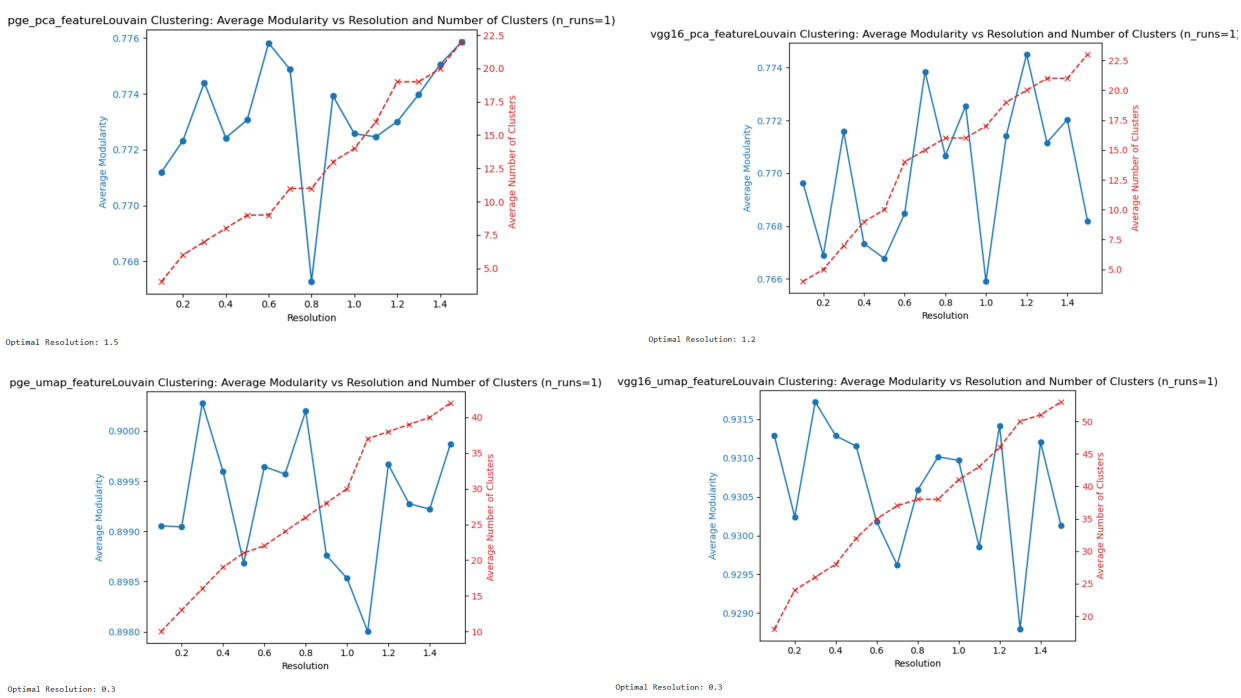



Figure 2.1.4.2: Plot of Average Modularity vs Resolution and Number of Clusters thought the PGA_PCA, VGG16_PCA, PGA_UMAP and VGG16_UMAP

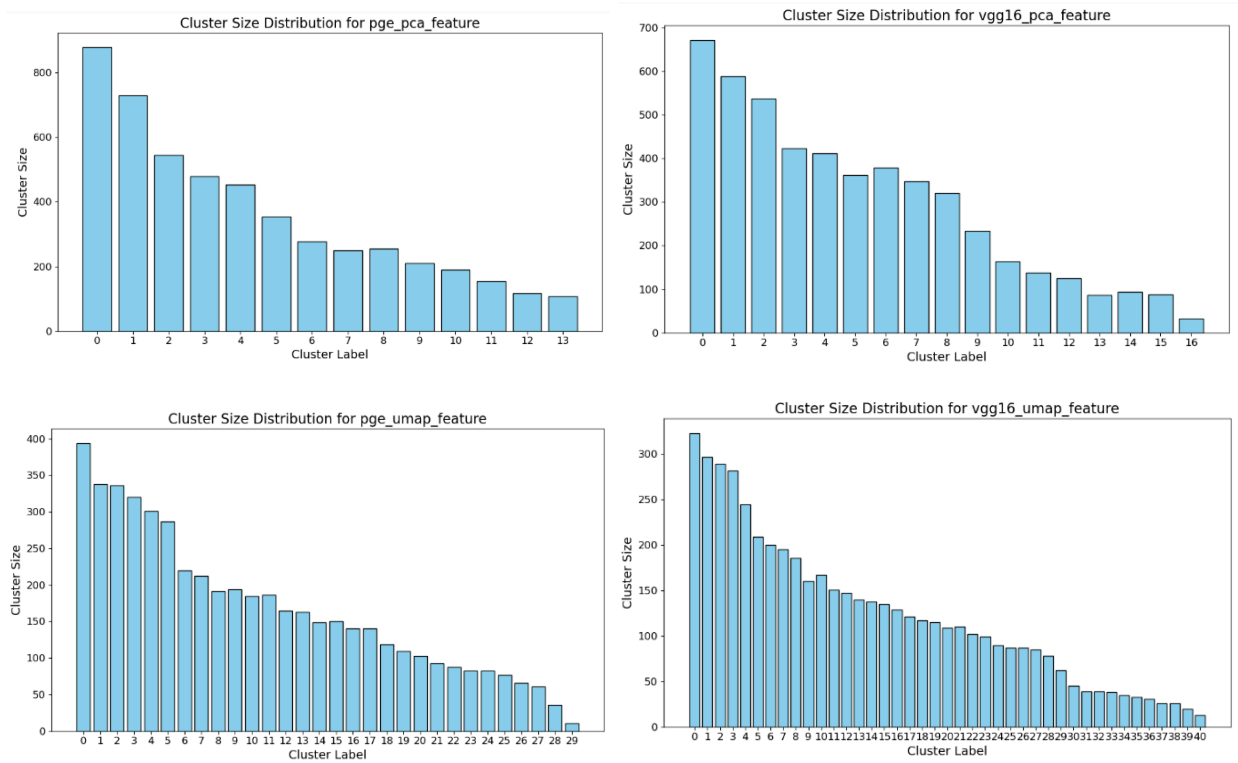



Figure 2.1.4.3: Cluster Size distribution for PGE_PCA, VGG16_PCA, PGE_UMAP and VGG16_UMAP

## 2.2 Evaluation Metrics

The Silhouette Score and V-Measure were used to evaluate the clustering performance of the four algorithms. The Silhouette Score measures how well a data point fits within its own cluster compared to others, with values ranging from −1 to +1, where higher values indicate better fit. We have the labels for the features. We also evaluate how well the clusters match the true labels using VMeasure. Which outputs values ranging from 0 to 1, where higher values indicate better alignment with the true labels.

# 3. RESULTS

## 3.1 Experiment Setup

As clustering is an unsupervised task, the entire dataset was used for clustering without splitting it into train and test sets. The grid search was performed on the following parameters given below.

| Algorithm | Representation | Parameters |
|---|---|---|
| K-Means | PathologyGAN | n_clusters, Init, N_init , Algorithm |
| GMM | PathologyGAN | Covariance_type, 'spherical', N_init, Max_iters, Tol, Init_params |
| DBSCAN | PathologyGAN | Eps and min_samples |
| Louvain | PathologyGAN | Resolutions, modularity and n_neighbors |

## 3.2 Experimental Results:

| Measure | | | | | | |
|---|---|---|---|---|---|---|
| **Representation** | **Feature Type** | **Evaluation Metrics** | **KMeans** | **GMM** | **DBScan** | **Louvain** |
| PathologyGAN | PCA | Silhouette | 0.258 | 0.005238 | 0.3285 | 0.138275 |
| | | VMeasure | 0.279 | 0.348642 | 0.3285 | 0.372472 |
| PathologyGAN | UMAP | Silhouette | 0.518 | 0.539330 | 0.5366 | 0.313317 |
| | | VMeasure | 0.337 | 0.559917 | 0.4913 | 0.356096 |
| VGG16 | PCA | Silhouette | 0.141 | 0.023029 | 0.342 | 0.082187 |
| | | VMeasure | 0.512 | 0.404575 | 0.256 | 0.637381 |
| VGG16 | UMAP | Silhouette | 0.668 | 0.533124 | 0.6739 | 0.408951 |
| | | VMeasure | 0.580 | 0.712210 | 0.5631 | 0.631743 |

## Discussion:

On one hand, in the case of PathologyGAN Features, Louvain and DBScan performed the best for both PCA and UMAP representations. This is likely due to compatibility of graph-base modularity optimization with non-linear relationships and density based approach. Kmeans performed moderately were not able to segregate clusters. Though GMM had an extremely weak Silhouette score, it had a strong V-Measure which implies GMM was able to handle cluster shapes better than cluster cohesion. As PathologyGAN is less compact and more noise-prone, DBScan’s density based clustering wasn't able to represent features.

On the other hand, in the case of VGG16 Features, Louvain and DBScan were able to perform consistently better than the other models. For PCA representation, KMeans and GMM struggled with overlapping clusters. But both were moderately able to align with the ground truths. For UMAP Representation, both KMeans and GMM performed extremely well in separating clusters and in alignment with ground truth. DBScan was able to create more meaning clusters as VGG16 captures more discriminative and hierarchical features.

## 4. CONCLUSION

To conclude, Kmeans simple clustering performed well on the UMAP feature sets of both PathologyGAN and VGG16. The Gaussian mixture model performed well on the UMAP feature set of the representations as well. Louvain clustering delivered the best results for VGG16 UMAP Feature. DB Scan performed well on both the representations with overall best scores. From the observed results, Graph-based clustering is more suitable for non-linear feature space like UMAPs. The combination of Feature representation like PGA and VGG16 and non-linear dimensionality reduction such as UMAP is more suitable for classification of tissue types compared to the other PGA.